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# Aditya Sneh

AI/ML Engineer

GitHub
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## Work Experience

**Project Junior Research Fellow** **Aug 2024 - Jul 2025**
*Systems and Informatics Research Laboratory* *Bhopal, MP*

- Built and deployed **LogMe**, a **24×7** Android (**Kotlin**) sensing platform that collected **250+ GB** of real-world mobile sensor and usage data from **70+ participants** over **14 days**, backed by **Apache/Python REST APIs** and **MySQL** for reliable ingestion.
- Engineered notification-triggered labeling workflows capturing **user activity**, **social context**, **intervention feasibility**, and **mood** for supervised learning and adaptive real-time modeling.
- Developed and **fine-tuned DySTAN** (Dynamic Cross-Stitch, Cross-Task Attention, BiLSTM), achieving **0.852/0.876 macro-F1** with **+21.8% improvement** over a CNN-BiLSTM-GRU baseline.
- Built end-to-end ML pipelines (segmentation, downsampling) for **production ML systems** and implemented a contextual multi-armed bandit (Thompson Sampling) for real-time optimization, exceeding the 0.5 intervention reward baseline.
- Collaborated with an AIIMS psychiatrist to integrate domain expertise into adaptive intervention policies; developed a **Streamlit dashboard** for real-time monitoring, **model evaluation**, and automated data validation.

**Gen AI Subject Matter Expert (SME)** **May 2024 - Jul 2024**
*Grades Buddy, Remote*

- Delivered **40+ production-grade GenAI solutions**, building and deploying **LLM-based** and generative models (GANs, diffusion) for support automation, document intelligence, healthcare support, and data-driven workflows.
- Built and **fine-tuned transformer-based architectures** (**LLaMA 2/3, Mistral 7B, BERT**) via Hugging Face, implementing sentiment analysis, contextual generation, and **RAG pipelines** achieving **85% task accuracy**.
- Applied **LoRA-based PEFT** for efficient domain adaptation and improved response reliability in **LLM systems**, including medical chatbot deployments.
- Developed a real-time speech-based sentiment classification system using **Whisper** for transcription and PyTorch classifiers, supporting scalable customer audio analysis.

## Skills

**Programming:** Python, SQL

**Machine Learning & Deep Learning:** PyTorch, Transformer-based Architectures (Hugging Face), scikit-learn, NLP, Computer Vision, Multimodal Learning

**LLM & Generative AI:** Large Language Models (LLMs), Fine-Tuning, LoRA/PEFT, Retrieval-Augmented Generation (RAG), OpenAI API, LangChain, LlamaIndex, Prompt Engineering, Agentic Systems

**Vector Search & Retrieval:** FAISS, Pinecone, Chroma

**Backend & Data Systems:** FastAPI, REST APIs, Pandas, NumPy

**Cloud & Infrastructure:** AWS, GCP, Docker

**MLOps & Production ML:** Model Deployment, Inference Optimization, CI/CD, MLflow, Monitoring & Observability, MLOps

**Version Control:** Git, GitHub

## Education

**BS-MS: Data Science & Engineering**, *Indian Institute of Science Education and Research (IISER), Bhopal* 2019–2024
CPI: 8.06/10

## Publications

- **Aditya Sneh**, A. Agarwal. **On the Robustness of Iris Presentation Attack Detectors**. BMVC 2025.
- **Aditya Sneh**, N. Sahu, A. Adyasha, A. Shelke, H. Lone. **HCFSLN: A Hyperbolic Few-Shot Learning Framework for Anxiety Detection**. arXiv 2025. Under review.
- **Aditya Sneh**, N. Sahu, H. Lone. **DySTAN: Joint Modeling of Sedentary Activity and Social Context from Smartphone Sensors**. arXiv 2025. Under review.
- A. Shelke, **Aditya Sneh**, A. Adyasha, H. Lone. **Fairness-Aware Few-Shot Learning for Audio-Visual Stress Detection**. arXiv 2025. Under review.
- N. Sahu, **Aditya Sneh**, H. Lone. **Real-World Receptivity to Adaptive Mental Health Interventions: Findings from an In-the-Wild Study**. arXiv 2025. Under review.

## Projects

### NotifSense: Real-Time Mobile Context Modeling & On-Device LLM Inference

- Built an on-device Android notification intelligence system in **Kotlin** that combines user context and notification metadata to make real-time **ALLOW / DELAY / SUPPRESS** decisions.
- Developed a **PyTorch** multi-label context model over **79** deployable mobile-sensor features; reduced the label space from **24** to **9** and tuned per-label thresholds, achieving **Macro F1 = 0.558** on held-out data.
- Optimized the HAR model for mobile with dynamic **INT8 quantization**, reducing size from **0.26 MB** to **0.08 MB** (**3.1x** smaller) while preserving on-device inference compatibility.
- Evaluated **TinyLlama-1.1B** for on-device reasoning using llama.cpp GGUF Q3/Q4 variants, enabling a privacy edge-LLM path under mobile memory constraints.

### SensorFusionAgent: Agentic AI for Research-Grade Sensor Harmonization

- Built a local-first async Agentic-AI platform for multimodal sensor harmonization that infers schema/task context, aligns IMU streams, and produces fused outputs with transparent quality, drift, and confidence reports.
- Implemented a **LangGraph** Planner–Executor–Observer loop for safe pre-fusion optimization, applying unit scaling, axis inversion, and smoothing only when quality improved by $\geq 0.002$.
- Designed **HQScore v4**, a 6-component fusion metric with explainable traces across **KL**, **Wasserstein**, **FFT/spectral similarity**, **cross-correlation**, **normalized DTW**, and **drift stability**.
- Added an adaptive ML layer with **GradientBoostingRegressor** models for sampling-rate and HQScore prediction, plus optional **gpt-4o-mini** fallback for schema inference on ambiguous inputs.

### Enterprise Decision Memory System (EDMS): Multimodal RAG Platform

- Engineered an enterprise RAG platform (**FastAPI + React/Vite**) for evidence-grounded Q&A over ADRs, RFCs, incident notes, postmortems, tickets, and architecture diagrams.
- Implemented a hybrid retrieval stack using **text-embedding-3-small**, **FAISS**, and **BM25**, with **gpt-4o-mini** generation constrained to retrieved enterprise evidence.
- Added multimodal ingestion with **gpt-4o** to convert uploaded architecture diagrams into retrievable text, plus **JWT-based RBAC** with **SQLite-backed authentication**.
- Developed an offline retrieval evaluation harness with **Precision@K**, **Hit@K**, and **MRR** over curated enterprise queries.